THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A model of a gas assumes particles have no volume. This is valid when: 1

(a) the gas is compressed to a liquid **(b)** the gas is cold
(c) the gas is at low pressure **(d)** the gas is solid

2. Assertion (A): A prediction is the same as a hypothesis. Reason (R): A prediction is a testable consequence drawn from a hypothesis. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The speed 3 x 10^8 m/s in km/h is about: 1

(a) 1.08 x 10^6 **(b)** 1.08 x 10^12 **(c)** 3 x 10^8 **(d)** 1.08 x 10^9

4. An area of 1 hectare equals 10000 square metres. A field of 2.5 hectares in square metres is: 1

(a) 250 **(b)** 25000 **(c)** 250000 **(d)** 2500

5. Assertion (A): Models of the atom have changed with time. Reason (R): New experiments gave results that older models could not explain. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A student discovers that her result of 9.78 m/s^2 for g differs from 9.81. The percentage difference is: 1

(a) 0.03 per cent **(b)** 3 per cent **(c)** 10 per cent **(d)** 0.3 per cent

7. If a laboratory scale has a least count of 0.01 g, the correct way to report a measurement is: 1

(a) 12.345 g **(b)** 12 g **(c)** 12.3 g **(d)** 12.34 g

8. Assertion (A): Weather forecasts for the next 10 days are less reliable than for the next 2 days. Reason (R): Small errors in the initial data grow as the forecast period increases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** A frictionless surface is used as an idealisation in many physics problems. 1
Reason (R): Idealisations remove less important details so that the main principle can be studied more easily.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The metre, the kilogram and the second are SI base units. 1
Reason (R): The newton is the SI unit of force.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State why the phrase 'a theory is only a theory' is a misunderstanding of science. 2

12. Convert 5.4 x 10^4 km/h to m/s. 2

13. State why peer review improves the quality of science. 2

14. What is meant by correlation not implying causation? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sample of metal weighs 27.0 g and has volume 10.0 cubic cm. Find its density in g/cm3 and kg/m3 and say which metal it might be. 3

16. Explain the importance of a pilot study in a scientific project with an example. 3

17. Explain how conflicting data from two labs can be handled. 3

OR

Explain how the need for a control group is illustrated by a test of a new tonic on plant growth. 3

18. Explain why the model of an ideal pendulum fails for large angles of swing. 3

OR

Describe how Indian scientists contributed to the progress of science with two examples (e.g. Saha, Raman, Bose, Aryabhata). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In the chemistry laboratory, Ms Nair shows the class a ball-and-stick model of a water molecule, with red and white balls joined by sticks. A student points out that real atoms are not coloured balls and bonds are not sticks. The teacher explains that the model still helps in predicting properties. 5
(i) What is a model in science?
(ii) Name one limitation of the ball-and-stick model.
(iii) Give one other example of a model used in science.
(iv) If an experiment gives a result that disagrees with the prediction of a model, what do scientists do, and why?

20. Analyse a data set showing that rainfall and crop yield are correlated across ten villages, and explain the steps required before claiming that rainfall causes high yield. 5

OR

Explain how mathematics provides the language for scientific laws by showing speed = distance/time, the use of graphs and units, and an example each of direct and inverse proportion. 5

END OF QUESTION PAPER — Total: 40 Marks